WORKED SOLUTION

STEP I 2009, Question 12

Probability and Statistics • Sampling without replacement

STEP I 2009, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q12

The key idea

Each probability here is a sum of products of bag counts divided by a power of the total, so every comparison comes down to comparing two polynomials in a, b, c. The opening inequality $x^2 + y^2 \geq 2xy$ is exactly the tool for that. In part (ii), counting "exactly two the same" head-on is fiddly. It is much easier to count the other two cases (all the same, all different) and subtract from 1.

Solution

The inequality

For any real x and y,

$$x^2 + y^2 - 2xy = (x - y)^2 \geq 0,$$

since a real square is never negative. So $x^2 + y^2 \geq 2xy$, with equality exactly when $x = y$.

Part (i)

Each bag holds $a + b$ sweets, and the two picks are independent because they come from different bags. Let S be the event that the two sweets have the same colour and D the event that they differ. Then

$$\mathrm{P}(S) = \mathrm{P}(\text{red, red}) + \mathrm{P}(\text{blue, blue}) = \frac{a}{a+b} \cdot \frac{b}{a+b} + \frac{b}{a+b} \cdot \frac{a}{a+b} = \frac{2ab}{(a+b)^2},$$

$$\mathrm{P}(D) = \mathrm{P}(\text{red, blue}) + \mathrm{P}(\text{blue, red}) = \frac{a}{a+b} \cdot \frac{a}{a+b} + \frac{b}{a+b} \cdot \frac{b}{a+b} = \frac{a^2 + b^2}{(a+b)^2}.$$

The denominators are equal and positive, and $2ab \leq a^2 + b^2$ by the inequality above. Hence $\mathrm{P}(S) \leq \mathrm{P}(D)$, as required, with equality only when $a = b$.

Part (ii)

Write $n = a + b + c$, the number of sweets in each bag. The contents are:

	red	white	yellow
bag 1	a	b	c
bag 2	b	c	a
bag 3	c	a	b

The three picks are independent, so the probability of any particular colour from each bag (in order) is the product of the three counts divided by n^3. Let A be the event that all three sweets are the same colour, T the event that exactly two are, and E the event that all three differ. These three events cover every possibility and don't overlap.

All the same. Reading down each column of the table,

$$\mathrm{P}(A) = \frac{abc + bca + cab}{n^3} = \frac{3abc}{n^3}.$$

All different. The colours from bags 1, 2, 3 form one of the $3! = 6$ orderings of red, white, yellow:

$$\begin{array}{lll} \text{RWY: } a \cdot c \cdot b & \text{RYW: } a \cdot a \cdot a & \text{WRY: } b \cdot b \cdot b \\ \text{WYR: } b \cdot a \cdot c & \text{YRW: } c \cdot b \cdot a & \text{YWR: } c \cdot c \cdot c \end{array}$$

So

$$\mathrm{P}(E) = \frac{a^3 + b^3 + c^3 + 3abc}{n^3}.$$

Ask yourself How do we get from these two answers to $\mathrm{P}(T)$ without listing the 18 remaining outcomes?

Since $\mathrm{P}(T) = 1 - \mathrm{P}(A) - \mathrm{P}(E)$, we need the expansion of $n^3 = (a + b + c)^3$. It has the three cubes, each of the six terms like a^2b with coefficient 3, and abc with coefficient 6:

$$(a + b + c)^3 = a^3 + b^3 + c^3 + 3\left(a^2b + b^2c + c^2a + ab^2 + bc^2 + ca^2\right) + 6abc.$$

Hence

$$\begin{aligned} \mathrm{P}(T) &= \frac{n^3 - 3abc - (a^3 + b^3 + c^3 + 3abc)}{n^3} \\ &= \frac{3\left(a^2b + b^2c + c^2a + ab^2 + bc^2 + ca^2\right)}{(a + b + c)^3}, \end{aligned}$$

which is the given result. The probability that the sweets are all the same colour is

$$\mathrm{P}(A) = \frac{3abc}{(a + b + c)^3}.$$

The deduction. Group the six terms in pairs that share a factor:

$$a^2b + b^2c + c^2a + ab^2 + bc^2 + ca^2 = a\left(b^2 + c^2\right) + b\left(c^2 + a^2\right) + c\left(a^2 + b^2\right).$$

By the opening inequality $b^2 + c^2 \geq 2bc$, and since $a > 0$, multiplying keeps the direction: $a(b^2 + c^2) \geq 2abc$. Similarly $b(c^2 + a^2) \geq 2abc$ and $c(a^2 + b^2) \geq 2abc$. Adding,

$$a^2b + b^2c + c^2a + ab^2 + bc^2 + ca^2 \geq 6abc.$$

Multiplying by $3/n^3 > 0$,

$$\mathrm{P}(T) \geq \frac{18abc}{n^3} = 6 \cdot \frac{3abc}{n^3} = 6\,\mathrm{P}(A),$$

as required. Equality needs $b = c$, $c = a$ and $a = b$ together, so it holds only when $a = b = c$.

Checks

- With $a = b = c$ every bag is one third of each colour, so $\mathrm{P}(A) = 3 \cdot \frac{1}{27} = \frac{1}{9}$ and $\mathrm{P}(E) = \frac{6}{27} = \frac{2}{9}$, leaving $\mathrm{P}(T) = \frac{2}{3} = 6 \cdot \frac{1}{9}$: the equality case.
- With $a, b, c = 1, 2, 3$ ($n = 6$): $\mathrm{P}(A) = \frac{18}{216} = \frac{1}{12}$, $\mathrm{P}(E) = \frac{54}{216} = \frac{1}{4}$ and $\mathrm{P}(T) = \frac{144}{216} = \frac{2}{3}$. These add to 1, and $\frac{2}{3}$ is 8 times $\frac{1}{12}$.
- We checked all three formulas by exact enumeration for every a, b, c from 1 to 8, and by simulation (for $a, b, c = 2, 3, 5$, 200 000 trials gave 0.659 against the exact 0.66 for $\mathrm{P}(T)$).

Common slips

- Writing the bag contents in the wrong order. Bag 2 has *a yellow*, not *a* red; a table like the one above prevents this.
- Assuming every ordering in $\mathrm{P}(E)$ gives abc. Only three do; the orderings RYW, WRY and YWR give a^3, b^3 and c^3.
- Forgetting to say that $a, b, c > 0$ when multiplying an inequality by a, b or c.

Beyond STEP

The final inequality is a case of Muirhead's inequality, a general rule for comparing symmetric sums of monomials; here it says the sum of the terms a^2b dominates the sum of the terms abc. Part (i) is the two-variable case of the same idea, and both rest on the AM–GM inequality. For more on counting arguments and complementary events, see Blitzstein and Hwang, *Introduction to Probability*.

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